

* SYMBOLIZATION HELP *

The HELP facility is *not* intended to provide help with the substance of logic. You can search this document to FIND a particular word or phrase by pressing Ctr L+F.

LOGIC NOTES:

- 1) Although there is a fairly broad consensus among logic teachers as to what constitutes a correct answer to a symbolization problem, certain fundamental differences between the syntax and vocabulary of natural languages and the syntax and vocabulary of our symbolic language have the effect of making symbolization an inexact science. At the present time, the program recognizes only one correct answer for most problems. If you believe that you have entered a correct answer that the program does not recognize, please send it us through the Feedback facility.
- 2) In order to constrain what is counted as a correct answer, we have adopted (by stipulation) certain conventions, many of which are traditional in logic texts (which is not to say that they are correct by more searching standards). In some, *but not all*, cases a comment will appear along with the problem reminding you of the convention. There is a thorough discussion of these conventions, along with examples and other aids to symbolization in the document, "Symbolization Examples and Explanations", invoked by the Examples button on the module's button bar. The program also includes exercises marked "EG", which exemplify solutions to certain problems.
- 3) Another source of help with symbolizations lies in the choice: "Hint", on the drop-down list. Hints provide the next step in the symbolization.
- 4) The program will inform you when you have entered an answer that is tautologically equivalent to a correct answer. In most cases, these answers are counted as correct. Your instructor can enable or disable this feature.
- 5) Each chapter of the TerryText contains an extensive discussion of symbolization.

Quick Start

Use the [Select button](#) to select a problem for symbolization. The Scheme of Abbreviation should appear in the left hand window (which can be resized by dragging the border). *The program is not able to check User problems that are not in its database.*

The standard method for symbolization is to build a tree in the main window (see [To use the tree window](#) below). However, if you believe that you can see immediately

what the symbolization should be, you can simply type the answer directly into the Direct Entry box, which is invoked by the [Direct button](#).

Check your work by clicking the [Check button](#) or pressing **Ctrl+k**. “Correct”, “Incorrect” or “Incomplete” appears in the upper right. The program also checks for answers that are tautologically equivalent to a correct answer. Unless your Instructor has chosen to treat all such answers as incorrect, “Correct Equivalent” appears in the upper right. In quantifier logic, the program cannot recognize *all* equivalents, but it does recognize many of them.

Print your work by clicking the [Print button](#) or pressing **Ctrl+p**. Use **Ctrl+Click** to select (or deselect) more than one exercise to print. You can either Print your complete work or Print Results, in which case only the problem plus “C” (for Correct) or “I” (for Incorrect or Incomplete) will be printed.

You can also submit your work over the Internet to a database of homework submissions established for this course. Click the [Submit button](#) at the bottom of the screen. You will be asked to verify course information, and to enter a password, which you will create the first time you make a submission. Use **Ctrl+Click** to select (or deselect) more than one exercise to submit. The program will verify that the submitted work has been received.

Save your work by clicking the [Save button](#) or pressing **Ctrl+s**.

Do not fear the [Delete button](#), it always first asks you to *confirm* what you wish to delete.

To use the tree window: Determine the primary logical structure (i.e. what the main logical connective would be) of the English sentence. Put the cursor in the text box containing the sentence, and right-click the mouse or press **Enter** to reveal a drop-down list of choices. (You can also use the key combination for the corresponding symbol, e.g. **Ctrl+Shift+C** if you believe the primary logical structure is that of a conditional.) Click on your selection, and a new set of text boxes will appear connected by the connective you have selected. Fill in the English content of the new text boxes, and repeat the process for each box. Ultimately, the English contents will have no further logical structure; they will be Atomic formulas. Select “Atomic” from the drop-down list, and fill in the symbolic formula in the pop-up window that appears. (The pop-up window is titled “Atomic Expression”).

IMPORTANT NOTE: the symbolizations of atomic formulas MUST be entered through the “Atomic Expression” pop-up window; they will not be recognized if entered directly into a text box.

Click the Check button to check your work or press **Ctrl+k**.

Error messages appear at the erroneous nodes. They are buttons; click them for an explanation. The explanation includes the proper content of the edit box from which the error was made, and the proper logical connective. The explanation window also contains simple ways to correct errors, and an Up button, which provides information about the node immediately above. In addition, the drop-down list contains the choice: “Hint”, which, if selected, also opens an explanation window containing a hint as to the next step. The information in these windows should be sufficient to explain, step by step, how to symbolize the given sentence of English.

Tools for the tree window: In addition to undo (**Ctrl+Z**) and the usual copy, cut, and paste, which are on the drop-down list, the module contains an Auto-copy facility, which allows you to load the copy buffer with the contents of a given text box simply by holding down the Alt key while invoking the drop-down list (e.g. by a right-click of the mouse). You can then right-click in the new edit box and paste that content into box, from which you can do additional editing to place the proper English sentence in the text box. **Alt+ArrowKey** moves you through the text boxes. The Checker only checks logical structure; it does not check the contents of the edit boxes, so you need only fill in what is needed to determine the structure of the succeeding node. If the tree becomes unwieldy because it is too wide, **Ctrl+Enter** will put an invisible ‘new line’ character into any text box, causing the formula to wrap at the insertion point.

Caveat: A bug in the underlying JAVA language sometimes prevents a text box from fully opening to accommodate pasted text. Press **Delete** or the **Spacebar** or click anywhere outside the field to remedy this. Indeed, clicking outside a text box will often correct a JAVA screen-writing error. In this, as in the other modules, bugs in the underlying JAVA language sometimes cause windows to be sticky, that is, for pieces of them to remain when they should be cleared away. Anything that causes the window to be redrawn will usually remedy the problem. The easiest method is to double click the title bar to maximize the window, and then to double click it again to restore it to normal size.

To use the Direct Entry box: Click the **Direct button** to invoke the box. Enter what you believe to be the correct symbolization, and click OK. Your symbolization will appear in the tree window in the form of a tree, with empty text boxes. Checking will now have the same effect it would have if you had built the tree directly in the tree window using the drop-down list, i.e. error messages will appear at the appropriate node on buttons that evoke explanations.

Tools for the Direct Entry box: A right-click in the Direct Entry box will invoke a Toolbox of logical symbols. Just click the symbol you wish to appear. You can also type through the Toolbox while it is open. Pressing Enter is equivalent to clicking the OK button. The standard Ctrl+Shift combinations for typing logical symbols are also available. They are shown on the hover messages that appear when the mouse is used to point at a Toolbox button.

Ctrl+Shift+C	→	(Conditional)
Ctrl+Shift+B	↔	(Biconditional)
Ctrl+Shift+A	^	(And)
Ctrl+Shift+O	∨	(Or)
Ctrl+Shift+N	~	(Negation)
Ctrl+Shift+E	∃	(Existential quantifier)
Ctrl+Shift+U	∀	(Universal quantifier)
Ctrl+Shift+I	≠	(Inequality)
Ctrl+Shift+D	ι	(Definite description operator)

Widen Window or Ctrl+Enter to Wrap

If the cursor is in an edit box, **Ctrl+Enter** will enter a 'new line' character, and cause the contents to wrap at the cursor. In this way, you can prevent the tree from scrolling off the screen.

When using the tree window to symbolize long sentences, it is often helpful to drag the sides of the window as wide as possible, and to drag the border between the tree window and the scheme of abbreviation window further to the left in order to maximize the width of the tree window. When necessary, scroll bars will appear on the tree window.

TABLE OF CONTENTS

Click on a topic

(1) The Symbolization Screen.

The screen of the Symbolization Module is divided into two large windows: the Scheme Window and the Tree Window, with a Results Bar along the top, and a Button Bar along the bottom.

(2) Selecting a problem to work on.

Click the “Select” button in the Button Bar at the bottom of the screen. This will bring up a list of exercises to choose from with the preceding choice highlighted. (If you are currently working on a problem, you will be prompted to save your work.) Exercises not yet attempted appear in **BLACK**. Exercises done correctly appear in **GREEN**. Exercises done incorrectly appear in **RED**. To select an exercise to work on, highlight it, and press **Enter**, click **OK**, or simply double-click the exercise. To move through the problem list use the **UpArrow**, **DownArrow**, **PageUp**, and **PageDown** keys. To jump to the top or bottom of the list use the **Home** and **End** keys.

The Symbolization Module cannot correct User problems that are not in its database.

(3) Building a symbolization tree.

The process of symbolizing proceeds a step at a time. At the start, you are given an English sentence to symbolize in a text box at the top of the Tree Window. Press **Enter** (or right-click with the mouse) to reveal a drop-down list of choices for the overall logical form of the English expression in the text box. When you select one of the forms (negation, conditional, etc.) and pressing **Enter** or double-clicking on your choice, new nodes of the tree will appear immediately below the previous level, showing the connective you have selected and providing a text box for each of the components of the connective (one for negation, two for conditional, etc.). Each new text box should then be filled in with the proper English expression. When you are satisfied that you have correctly filled in a text box, press **Enter** (or right-click the mouse), and repeat the process. If the English expression in the text box contains no further logical structure, as represented by a symbolic connective, select “Atomic” from the drop-down list, and you will be presented with a pop-up window, titled “Atomic Expression” allowing you to type in the symbolic formula (e.g. P , Q , Fx , Fy , etc.) that abbreviates the English expression. *Note that the program will not recognize such formulas unless they are entered by means of the “Atomic Expression” pop-up window.* In providing atomic symbolic formulas, you should be guided by the Scheme of Abbreviation, which appears in the Scheme Window.

Continue the process of filling in text boxes and pressing Enter until all the branches terminate in 'Atomic Expressions'. At this stage, you will have completed your symbolization.

When you select a quantifier from the drop-down list, the program will automatically select the generalized variable in such a way that a quantifier that lies within the scope of another quantifier will always have a different generalized variable than the quantifier within whose scope it lies. This helps to avoid errors. A similar practice should be adopted for manual symbolizations.

Revising a symbolization tree

If you wish to revise a prior choice from the drop-down list, put the cursor into the edit box immediately above it. You can now make a new choice from the drop-down list. If you correct your choice of connective, the tree below remains. To delete the tree below a given node, select "None" from the drop-down list. If you revise a two-place connective to a one-place one (e.g. conjunction to negation) the right branch below will be deleted.

For quantifier logic, the choices: Restricted Universal Generalization (Rest Univ) and Restricted Existential Generalization (Rest Exist) produce, respectively, a universal generalization of a conditional and an existential generalization of a conjunction. In correcting an error, changing a conditional node to a Restricted Universal Generalization will insert a Universal Generalization above the node. Changing a conjunction node to a Restricted Existential Generalization will insert a Existential Generalization above the node.

The Resulting expression

Each time you add new nodes to the tree by selecting a connective from the drop-down list the connective is incorporated into the expression in the Results Bar at the top of the screen. Also, each time you complete filling in a text box, and leave it, the contents of the field are also incorporated into the expression in the Results Bar. At each stage in the development of the tree, the Results Bar will display the current stage of the symbolization (the current Result). When you have completed your symbolization, the Result, displayed in the Results Bar, should be a symbolic sentence.

Symbolization shortcuts

A blank text box will leave a blank in the corresponding position in the expression in the Results Bar. However, since all English expressions in text boxes are ultimately replaced by symbolic expressions, it is possible to build the tree filling in only those text boxes required to aid your memory as to the stage of the symbolization. Thus if you can see some (or all) of the logical structure at a glance, without the aid of the intermediate English formulations, you can complete the

symbolization just by using the usual **Ctrl+Shift+<key>** combinations to determine the main connectives, and filling in the atomic symbolic formulas using the drop-down list.

An equivalent method to build a tree quickly, though with empty edit boxes, is to use the Direct Entry box. The box is invoked by the Direct button at the bottom of the screen. Simply enter your symbolization into the box, and press Enter or click OK, and the equivalent tree will appear in the Tree Window (with all blank edit boxes). It can now be checked just as any symbolization tree would be. The Direct Entry box contains a Toolbox of symbols, invoked by a right-click of the mouse, and also responds to the same keystroke combinations as the Derivations module.

Ctrl+Shift+C	→	(Conditional)
Ctrl+Shift+B	↔	(Biconditional)
Ctrl+Shift+A	^	(And)
Ctrl+Shift+O	∨	(Or)
Ctrl+Shift+N	~	(Negation)
Ctrl+Shift+E	∃	(Existential quantifier)
Ctrl+Shift+U	∀	(Universal quantifier)
Ctrl+Shift+D	ι	(Definite description operator)

Navigation within the drop-down list

To select an item from the drop-down list, you may either highlight the item, and press Enter, or simply double click the item. You can highlight an item by clicking or using the Up and Down Arrow keys. Home, and End are also available. Typing an initial letter or sequence of letters, e.g. “a” for “Atomic”, will also cause the first item beginning with that letter or sequence of letters to be highlighted.

Editing shortcuts in edit fields

There are a number of editing shortcuts that may prove useful for filling in the text boxes.

The usual editing keys are available.

Ctrl+A	Highlight everything in the text box
Ctrl+C	Copy the highlighted material
Ctrl+X	Copy and delete the highlighted material
Ctrl+V	Paste the copied material at the cursor location
Ctrl+Z	Undo the most recent paste or delete
Alt+Enter	Wrap Edit field at the insertion point

The usual alternatives are also available

Ctrl+Insert	Copy the highlighted material
--------------------	--------------------------------------

Shift+Delete
Shift+Insert

Copy and delete the highlighted material
Paste the copied material at the cursor location

In addition, the module contains an Auto-copy facility. holding down the **Alt** key while either using the **Ctrl+Shift+<key>** combinations or while invoking the drop-down list (either by pressing **Enter** or by a right-click of the mouse) will cause the contents of the current edit box to be automatically copied to the copy buffer. The contents can then be pasted (**Ctrl+v**) into the edit boxes at the next node, and edited appropriately. This is often faster than retyping or other copying methods.

(4) Checking your work.

You can check your work at any time by clicking the “Check” button at the bottom of the Derivation screen (or by **Ctrl+k**). If your symbolization is correct, the message “Correct” will appear in the upper right hand corner as part of the Results Bar. Other possible messages are “Incomplete” and “Incorrect”.

The program checks only the logical structure of your current Result (the expression in the Results Bar). It does not attempt to check the English. Thus, if the logical structure, or partial logical structure appears to be correct, no error messages will appear. This can lead to confusion when the logical structure is correct, but the symbolization tree is not complete. For example, if the top node is correctly designated as a conditional, no error message will be displayed even if the English in the text boxes reverses antecedent and consequent. Of course, such an error will eventually be expressed by an incorrect choice of connectives or atomic formulas at lower nodes. However, until such lower nodes are entered into the symbolization, a check may falsely appear to confirm the entire tree when in fact it is only confirming the symbolic portion. To avoid such confusions, complete the symbolization before checking.

Error messages

If the tree contains an error, an Error Indicator, **Error**, will appear next to each highest-node *erroneous logical symbol*. A highest-node error is one with no errors above it. Should you be puzzled by such an Error Indicator, remember that the program is only checking logical structure; it is not checking the English expressions. So what appears to be correct, based on the English expression at the preceding node, may actually reflect an error in the English.

The Error Indicator is actually a button. Clicking an Error button will produce a message indicating what the correct logical symbol should have been, *and* what the correct English expression should have been at the preceding node. If this

information is not sufficient, an UP button will indicate what the correct English expression should have been at the node preceding that one. And so on.

The Error messages are not saved with the work, so you will also need to recheck your work when an incorrect or incomplete problem that has been saved is recalled.

(5) Saving your work.

Save your finished or unfinished work on a problem by clicking the “Save” button at the bottom of the Symbolization screen (or by **Ct r l+S**). This will save the current version of your work to your Work Disk. You can confirm that your work has been saved by clicking “Select” and re-selecting the problem. None of the Error messages are saved with the problem. So when you recall a prior problem, you will have to Check it again.

(6) Printing your work.

You can print your work at any time by clicking the Print button at the bottom of the Symbolization screen (or by **Ctrl+p**). The printout will contain your user information (name, Student ID#, etc.) as a header. (Note: At UCLA, you must first open a “printing account” at the lab desk.) At present only Results printing, which just prints a single line for each problem stating whether it was attempted, and if so whether it was completed correctly, is available for the Symbolization module.

(7) Submitting your work.

You can also submit your work over the Internet to a database of homework submissions established for this course. Click the Submit button at the bottom of the screen. You will be asked to verify course information, and to enter a password, which you will create the first time you make a submission. Use **Ctrl+Click** to select (or deselect) more than one exercise to submit. The program will verify that the submitted work has been received.

The database will match your submissions with Assignments constructed by your instructor. You may make multiple submissions of the same problem. The submission that “counts” is the last submission made before the due date of the assignment.

You can view the assignments, and your submissions to each assignment, by clicking “Assignments” on the Menu.